



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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Sign language translator

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Agenda

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- Introduction
- Existing System
- Proposed System
- System Architecture
- Requirements Specifications
- System Models
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Abstract

The **Sign Language Interpreter and Learning Management System** is a Python and MySQL-based application designed to reduce communication barriers by providing information about sign-language gestures and their meanings. The system allows users to search for signs, view translations, maintain translation history, access learning modules, and track their learning progress. MySQL is used to store and manage users, gestures, translations, learning modules, and progress records, while Python provides the application interface and logic. The project demonstrates important DBMS concepts such as relationships, normalization, SQL queries, joins, triggers, stored procedures, and report generation.

Introduction

Sign language is an important means of communication for deaf and hard-of-hearing individuals, but communication becomes difficult when others do not understand it. The proposed system provides a centralized platform where users can search and learn sign-language gestures along with their meanings. It uses Python for the application and MySQL for storing and retrieving information. The system aims to make sign-language learning more organized and accessible while demonstrating the practical application of database management concepts.

Existing System

Existing sign-language learning methods mainly depend on books, dictionaries, videos, websites, and general translation applications. These methods often lack centralized storage, personalized learning progress, translation history, and structured management of sign-language information. Users may have difficulty tracking the signs they have learned, while administrators may not have an efficient way to manage gesture and translation data. Therefore, there is a need for a centralized database-driven system that combines sign information with user and learning management.

Proposed System

The proposed system provides a centralized platform for managing and learning sign-language gestures. Users can register, log in, search for gestures, view their meanings, access learning modules, and maintain translation history and progress. Administrators can add, update, and delete gesture, translation, and learning-module records. Python handles the application interface and processing, while MySQL stores the system data and manages relationships between different entities, providing an efficient and organized solution.

Sign language references (image)



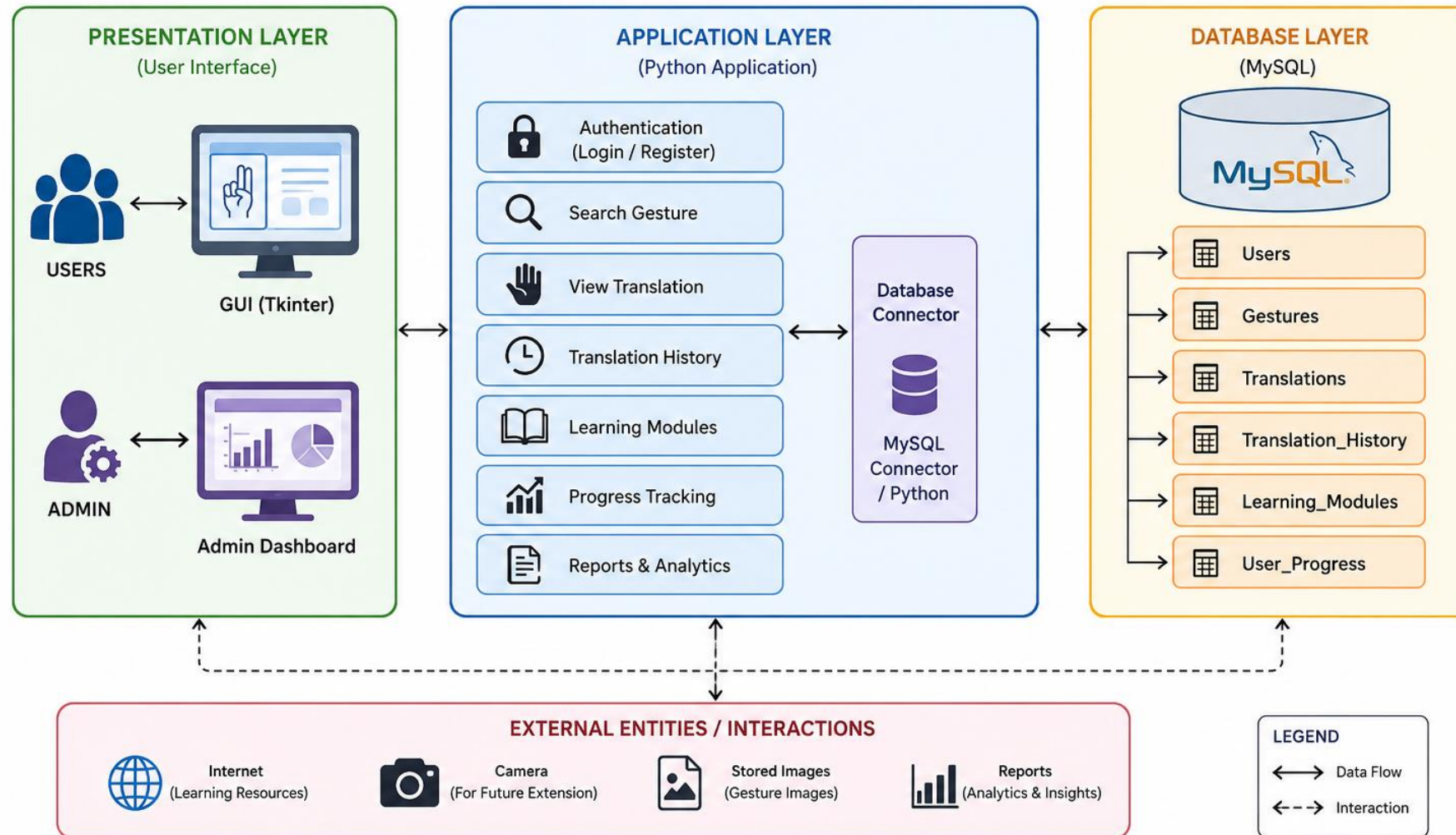
The project demonstrates DBMS concepts including *primary and foreign keys, normalization, constraints, joins, triggers, stored procedures, and report generation*.

System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and database layer. The presentation layer is developed using Python Tkinter and provides the user interface. The application layer uses Python to handle authentication, searching, translation retrieval, learning activities, and database operations. The database layer uses MySQL to store users, gestures, translations, translation history, learning modules, and user progress. Python connects to MySQL using MySQL Connector/Python.

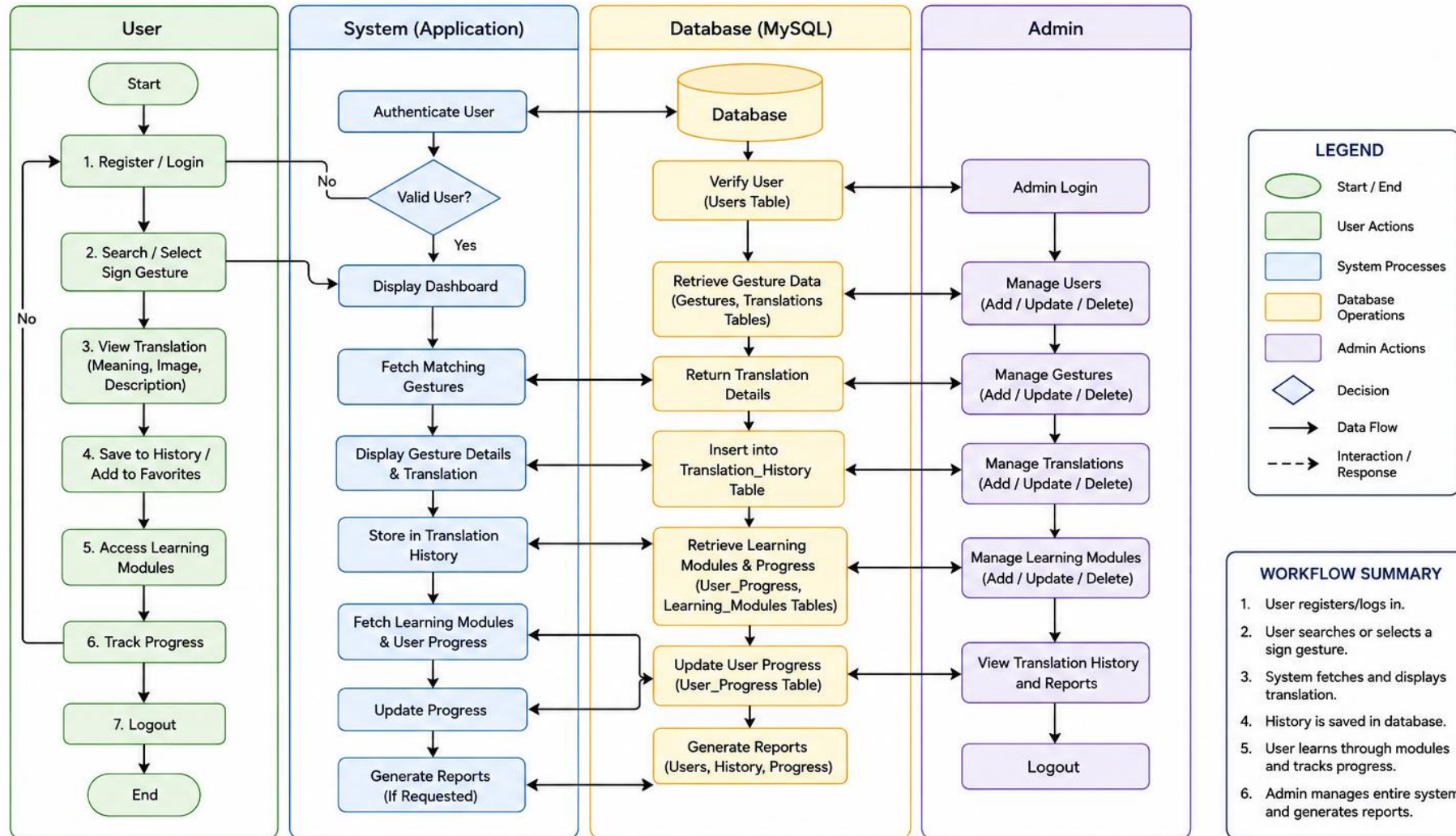
SYSTEM ARCHITECTURE DIAGRAM

Sign Language Interpreter and Learning Management System



WORKFLOW DIAGRAM

Sign Language Interpreter and Learning Management System



System Models

The system models describe the structure and interaction of the application using ER, relational, use-case, and data-flow models. The main entities include **Users, Gestures, Translations, Translation History, Learning Modules, and User Progress**. Relationships are established using primary and foreign keys. Users can search and view translations, maintain history, and track learning progress, while administrators manage system data. The models help represent how information is stored, related, processed, and exchanged within the system.

Requirements Specifications

The system requires a computer with at least 4 GB RAM, Python 3.x, MySQL Server, MySQL Workbench or XAMPP, Visual Studio Code, and MySQL Connector/Python. Functionally, the system should support user registration and login, gesture searching, translation display, translation-history management, learning modules, progress tracking, administrator management, and report generation. The system should also provide security, reliability, usability, good performance, and the ability to add new gestures and learning content in the future.

Conclusion

The **Sign Language Interpreter and Learning Management System** provides a practical database-based solution for organizing sign-language information and supporting users in learning signs. By using Python and MySQL, the system combines a simple user interface with efficient data storage and retrieval. It demonstrates essential DBMS concepts such as database design, normalization, relationships, SQL queries, joins, triggers, stored procedures, and reporting. The system can be further enhanced in the future by integrating computer vision and machine learning for automatic sign recognition.